BantayLaot+

ENHANCED FISHERIES MONITORING SYSTEM

An Enhanced System for Offline and Secure Monitoring of Philippine Municipal Fisheries with AI Integration Preparedness

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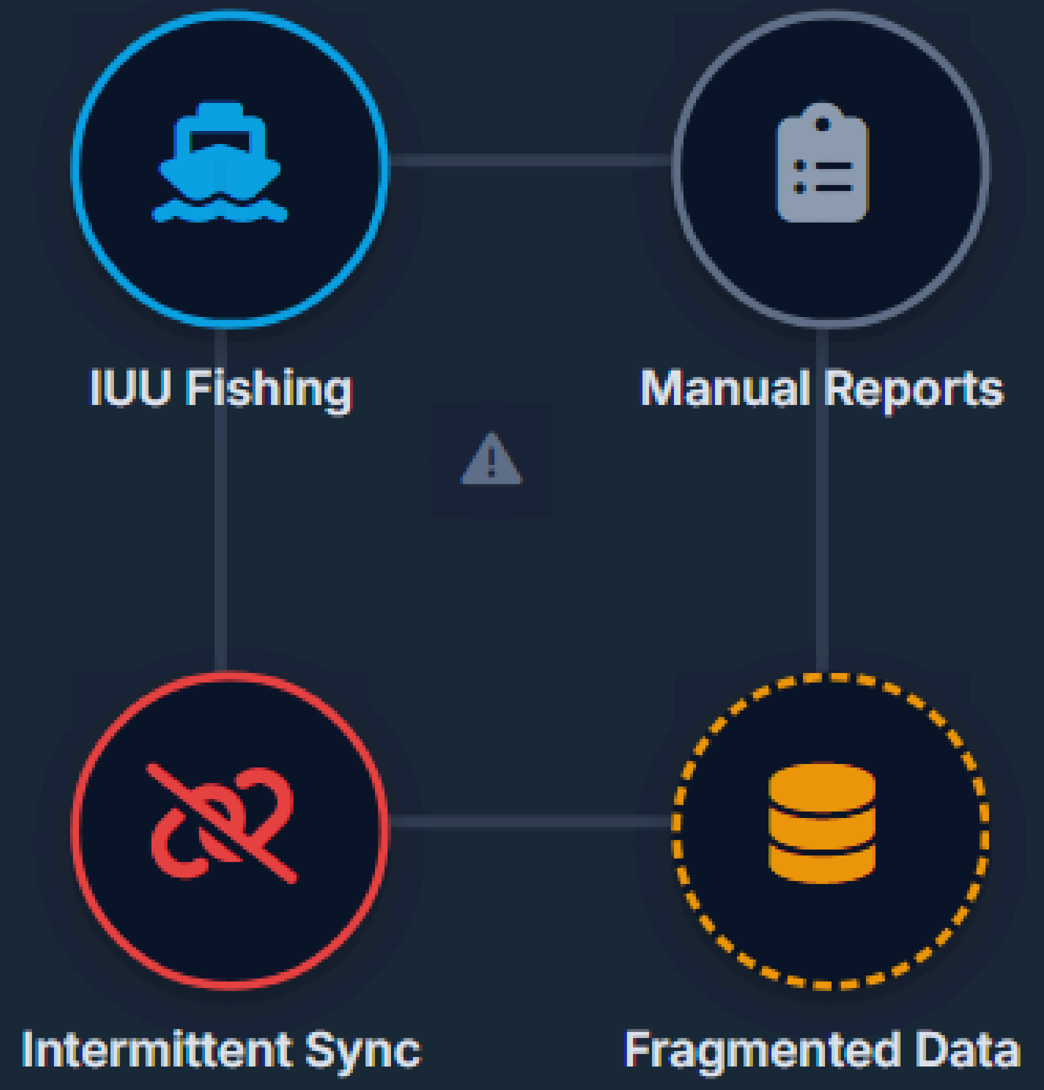
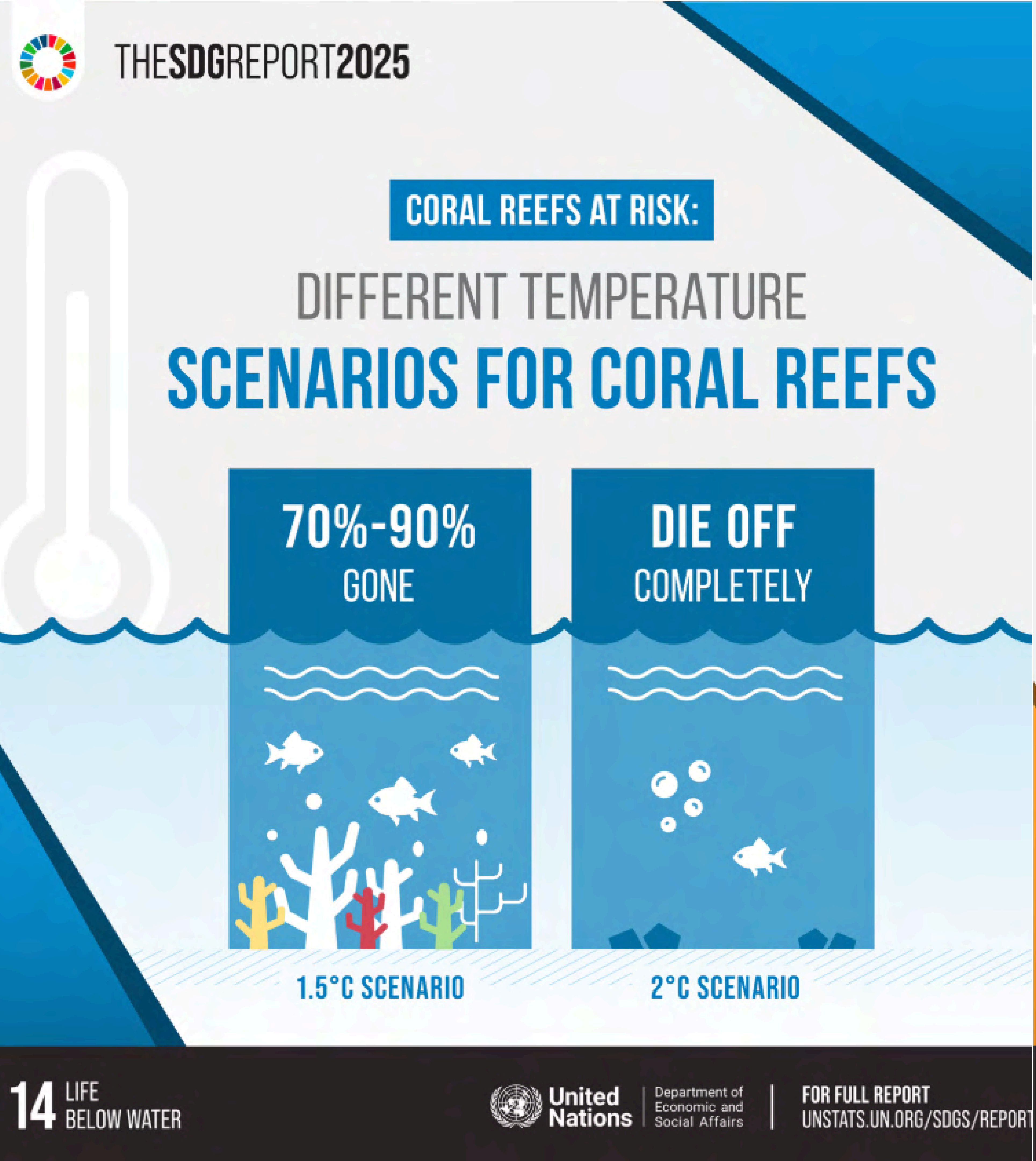
GPS Tracking | Offline-First | Secure | AI-Ready

The Problem with Municipal Fisheries Monitoring

- **IUU fishing** threatens coastal livelihoods and fish stock sustainability.
- Enforcement agencies rely heavily on **manual, paper-based** reporting.
- **Intermittent internet connectivity** in remote municipalities breaks standard digital tools.
- **Fragmented datasets** prevent cross-municipality oversight by central agencies and researchers.




THESDGREPORT2025
CORAL REEFS AT RISK:
DIFFERENT TEMPERATURE
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BantayLaot (Gallero, 2024): What Was Missing

Gap in Original System	Operational Impact
No account or authentication system	No session continuity; any device could access data
Single undifferentiated administrative role	Researchers, enforcers, and admins shared the same view
No structured sync state management	No way to know which records had reached the server
No organized image pipeline	Photos unstructured; unusable for future AI training
Municipal-only data silo	No mechanism for cross-municipality aggregation

BantayLaot+: Before → After

System improvements and enhanced features comparison

System Comparison: Original vs Enhanced

Before (BantayLaot) — Original System	After (BantayLaot+) — Enhanced System
No account or authentication system No session continuity; any device could access data	**JWT authentication; token persisted in SharedPreferences** NEW Works offline with secure session management
Single administrative role Researchers, enforcers, and admins shared the same view	**3 role-differentiated dashboards** ENHANCED Municipal, Central, Researcher roles
No structured sync state management No way to know which records had reached the server	**WorkManager: PENDING → SYNCED / FAILED** NEW Per record sync state management
No organized image pipeline Photos unstructured; unusable for future AI training	**Cloudinary pipeline with metadata** NEW Structured for future AI training
Municipal-only data silo No mechanism for cross-municipality aggregation	**Two-tier server architecture** ENHANCED Municipal (localhost) + Central (Railway) aggregation

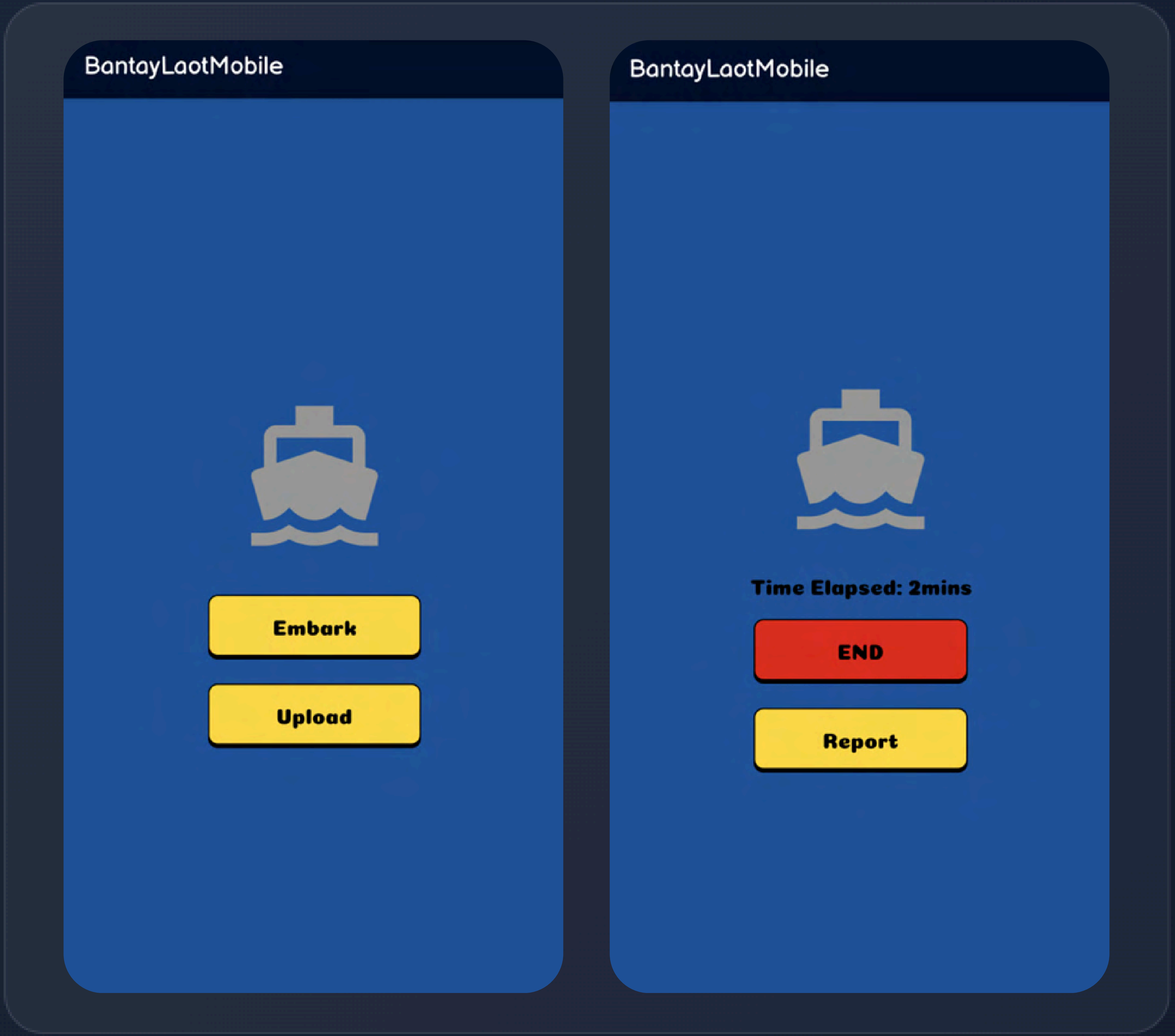
System Status: Fully Operational 5/5 improvements implemented

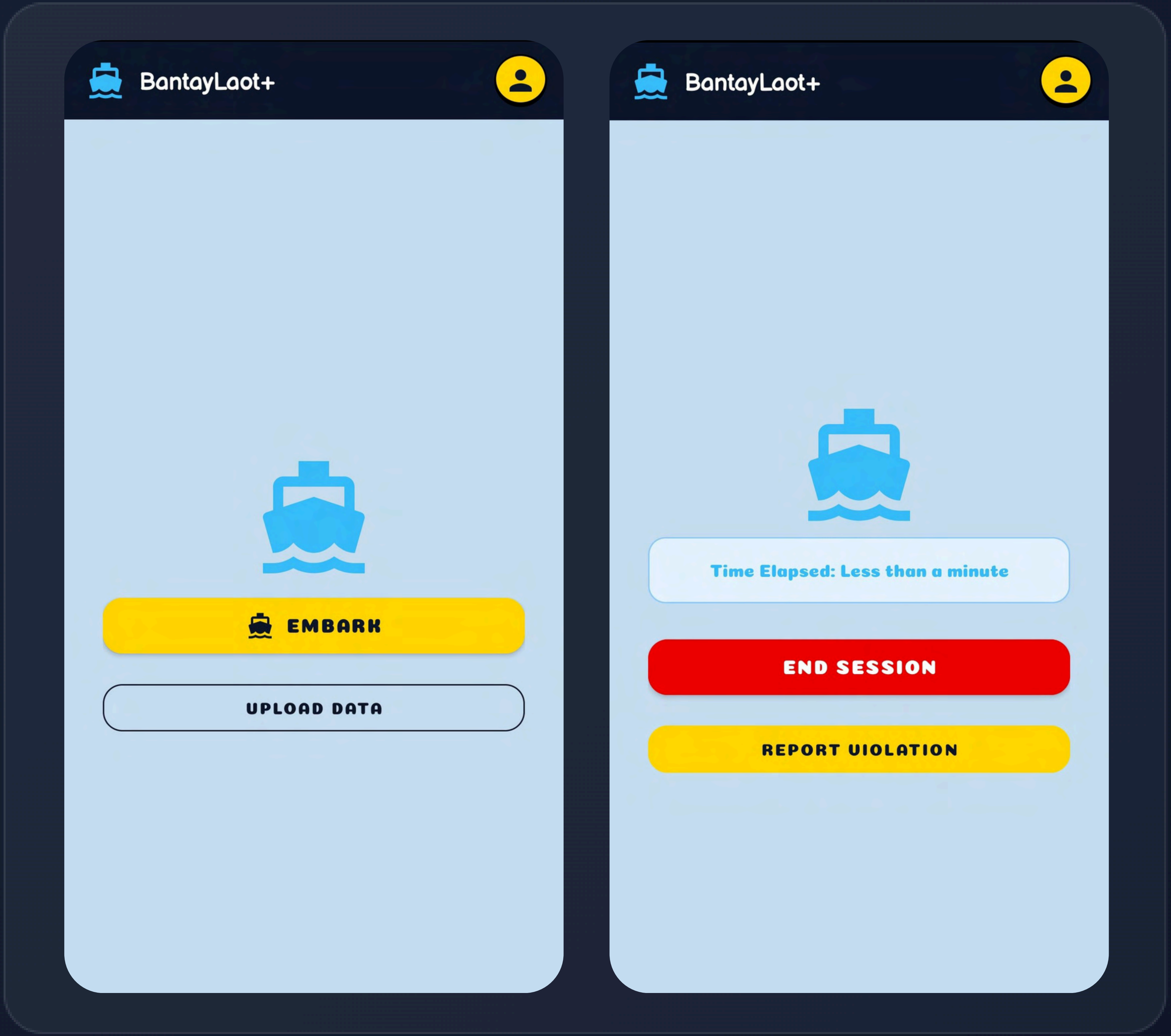
Last updated: May 2024

Mobile **App**

Before vs After

System Operational

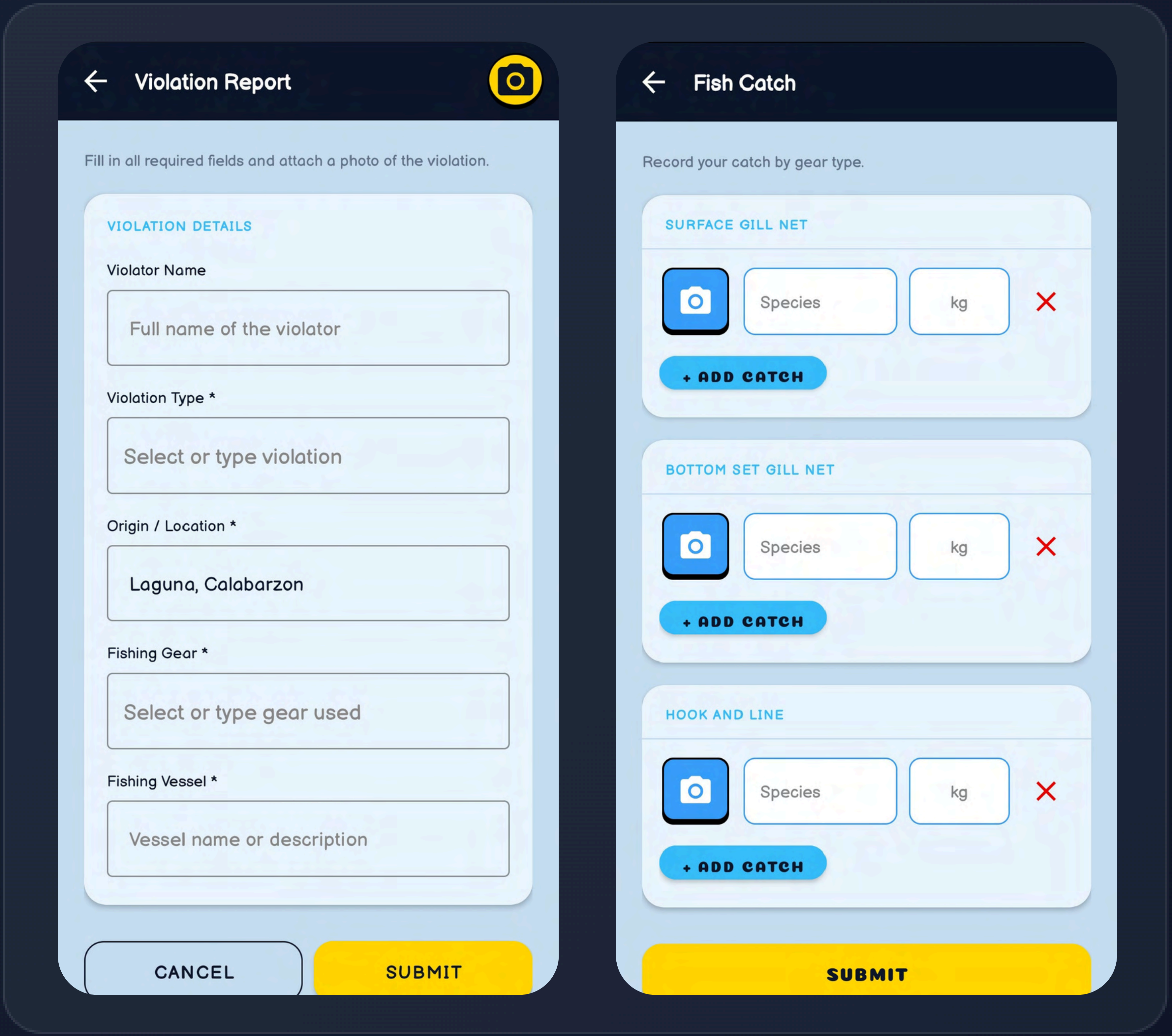




Mobile App

Before vs After

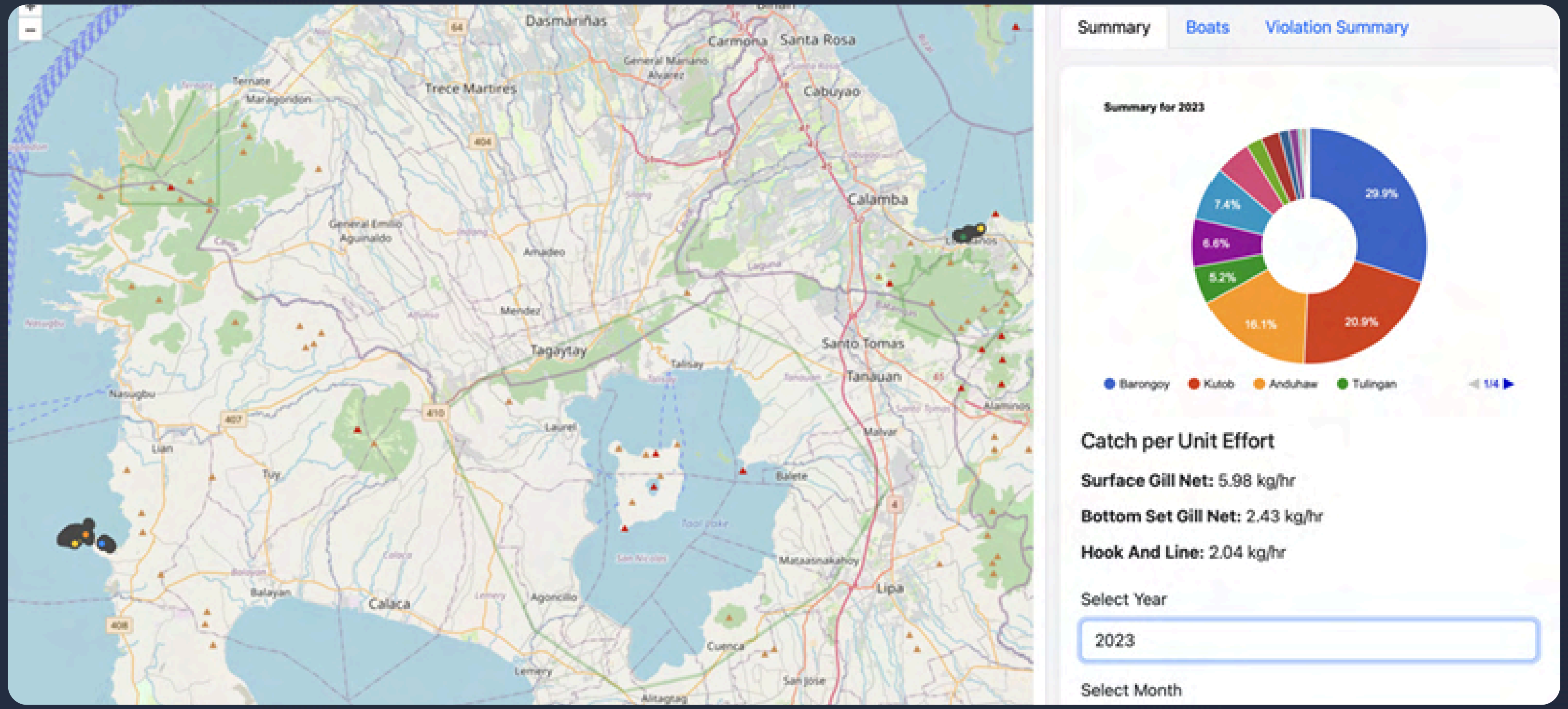
System Operational

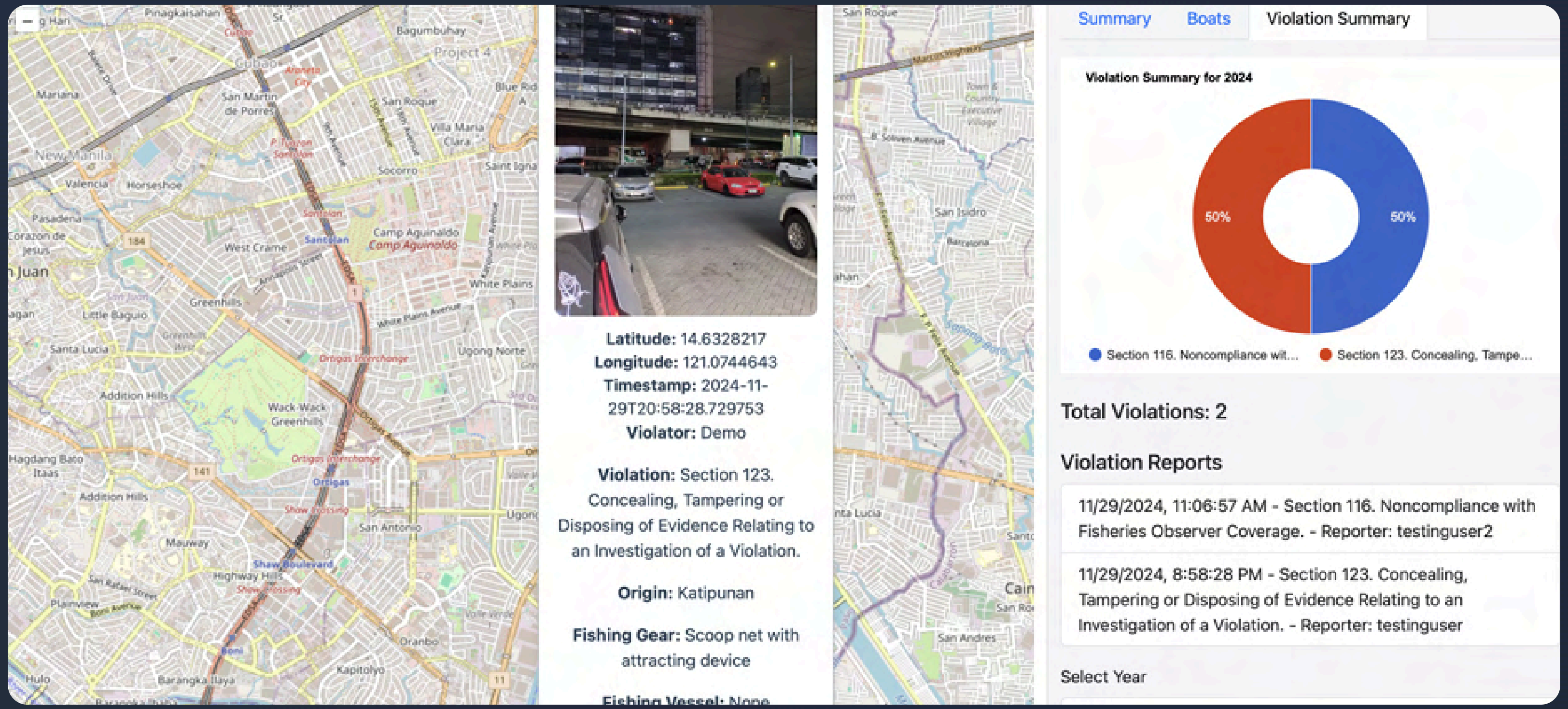


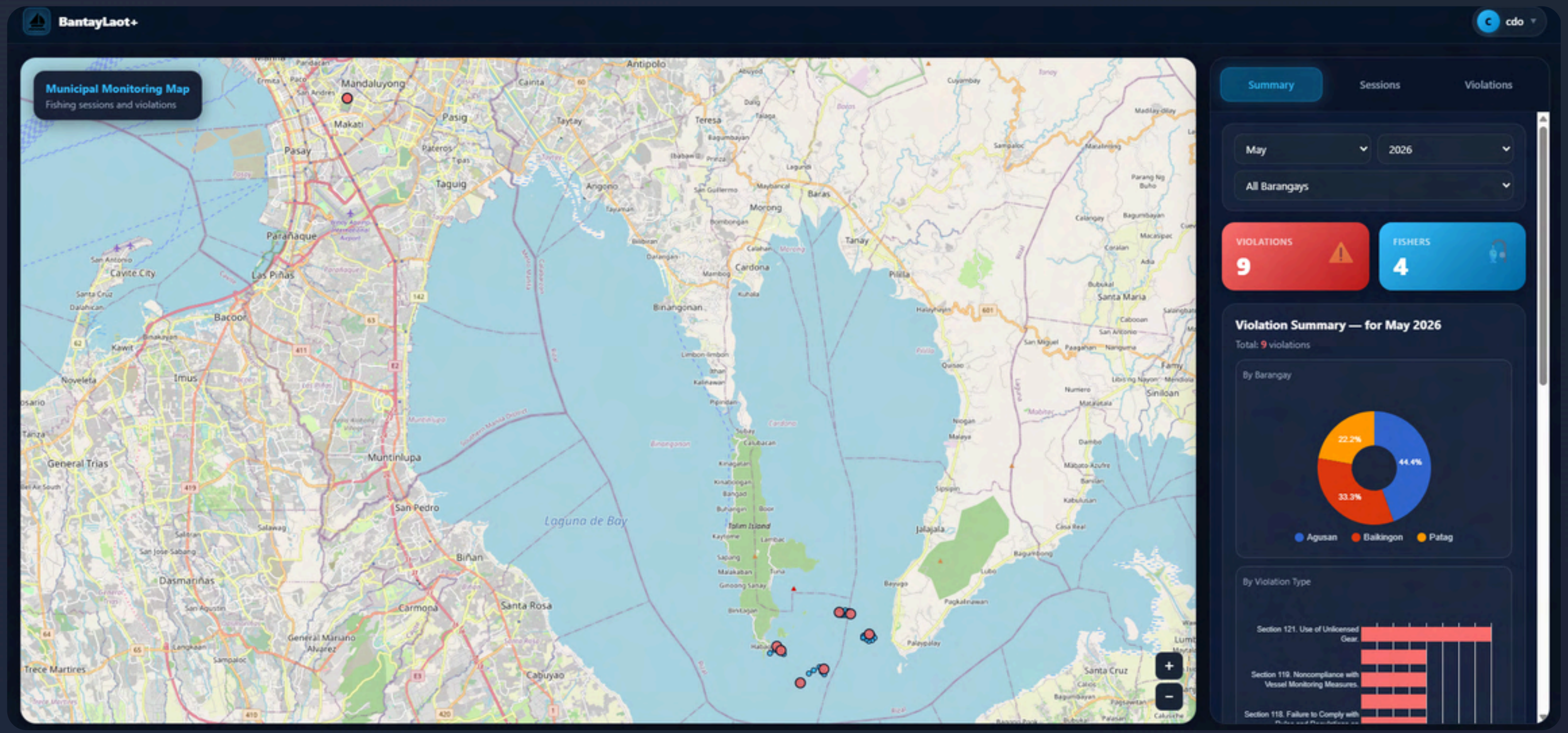
Web Dashboards

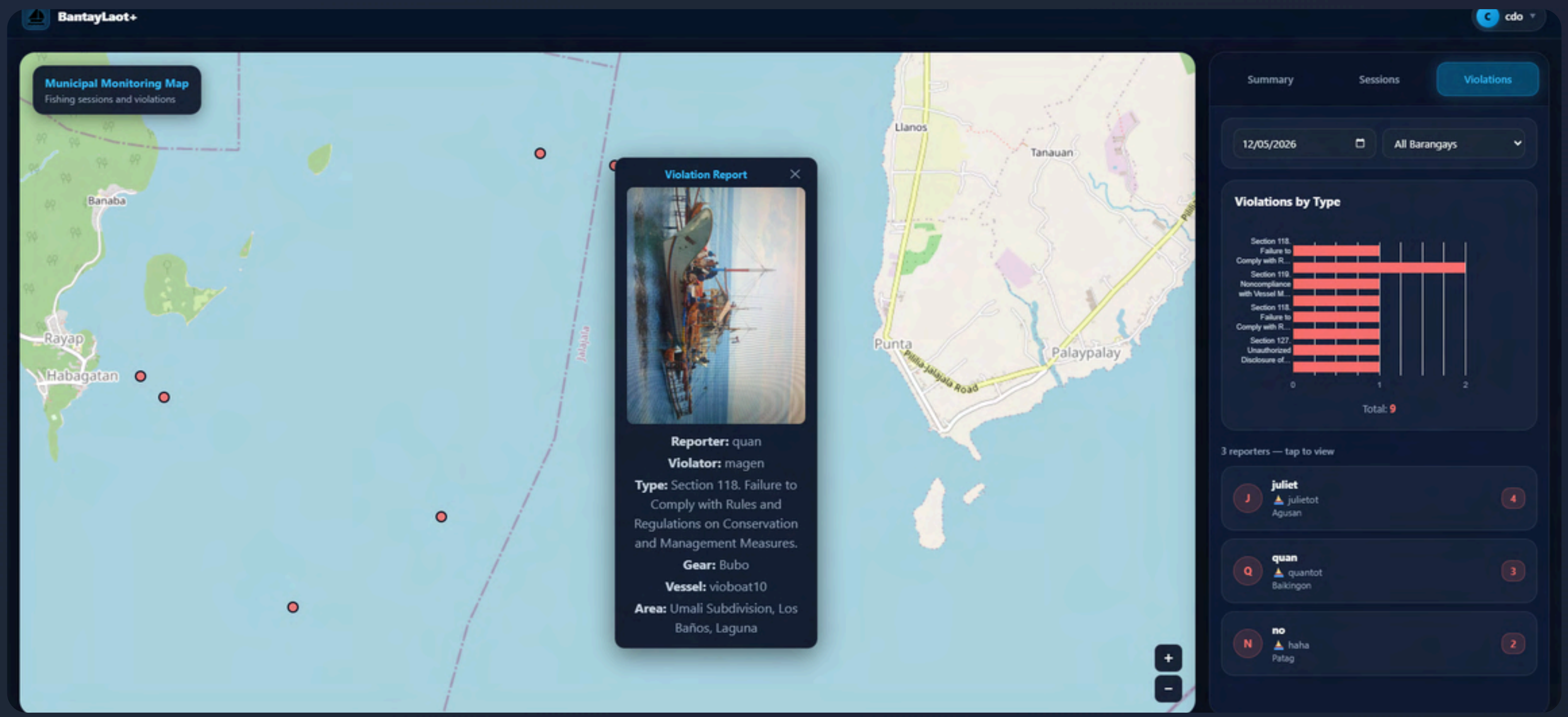
Before vs After

System Operational









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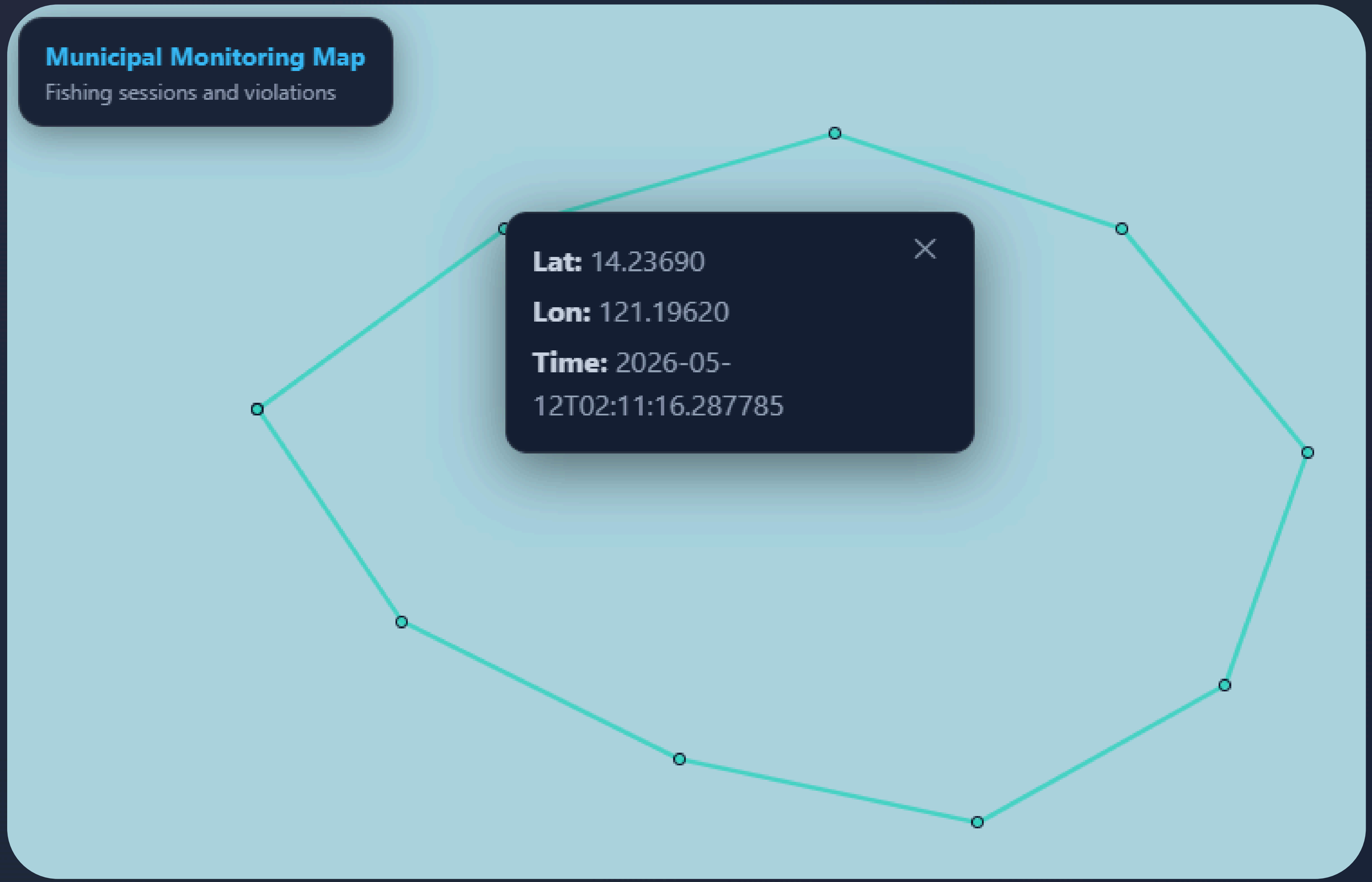


System **Architecture**

Multi-tiered architecture with offline-first design

MERN Stack | Room DB | WorkManager | JWT

Client Layer
Mobile & Web Apps

- **Android App** — Kotlin + Room DB
- **Web App** — React.js + Node.js
- **GPS Tracking** — Location Services

API Layer
RESTful Services

- **Municipal API** — Express.js + MongoDB
- **Central API** — Railway-hosted
- **Auth Service** — JWT + Middleware

Processing Layer
Data Processing

- **WorkManager** — Background Sync
- **JWT Persistence** — Token Management
- **Conflict Resolution** — Timestamp-based

Data Layer
Storage & Persistence

- **Room DB** — Local SQLite
- **MongoDB** — Municipal Data
- **Cloud Storage** — Railway + Cloudinary

System Status: **Operational** Client API Processing Data Uptime: **99.9%**

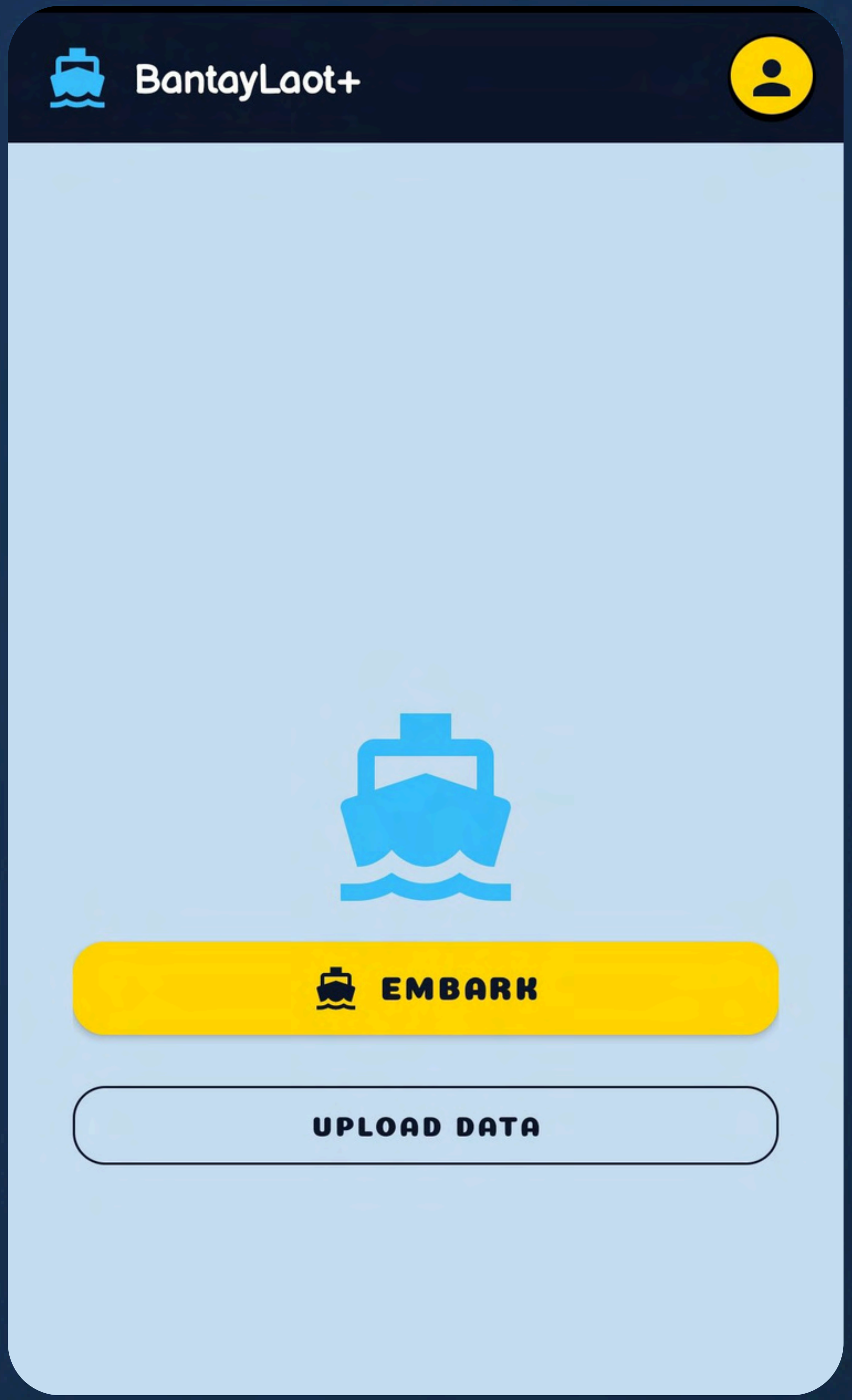
Mobile Application Workflow

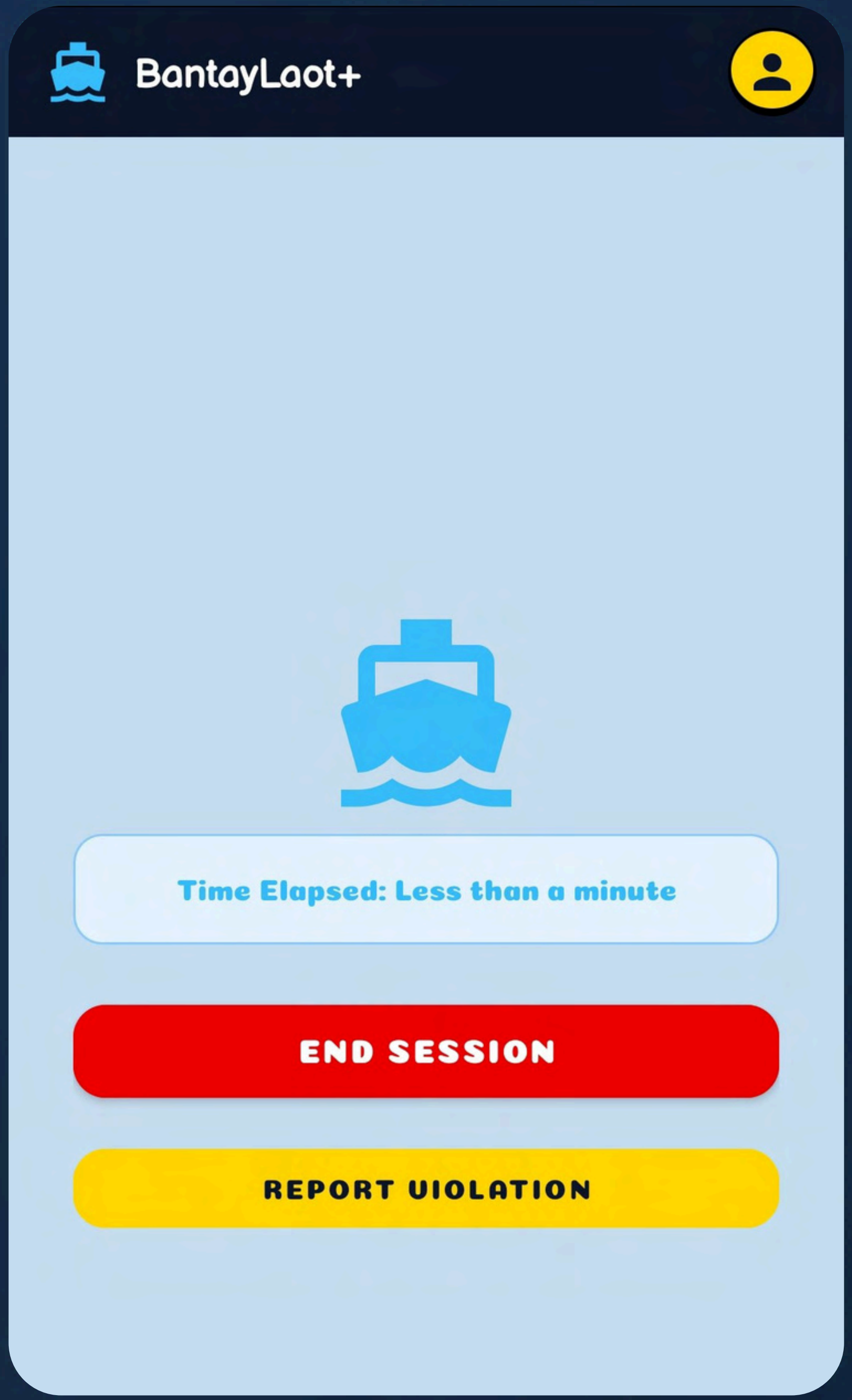
Four-stage fishing session management

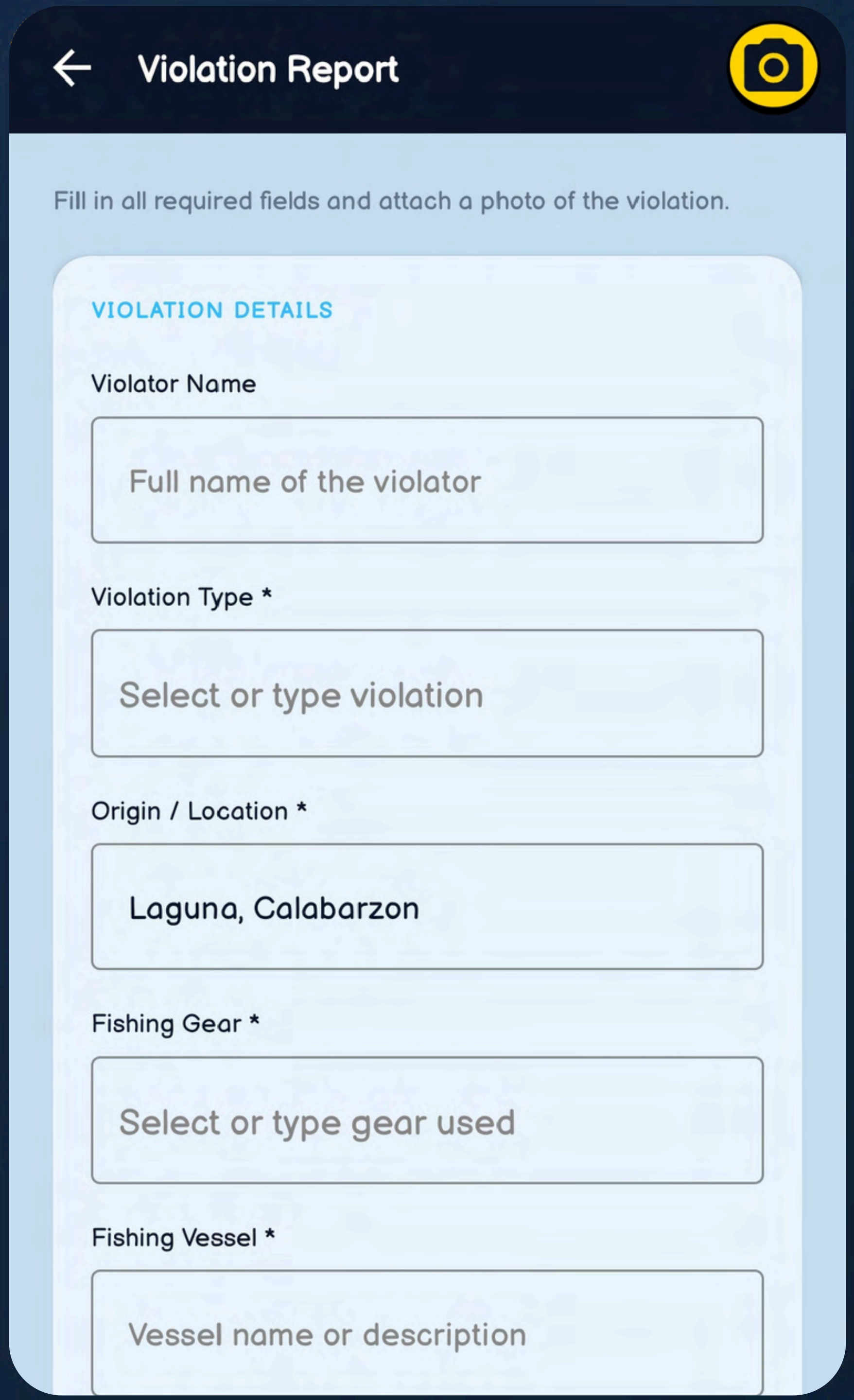
System Active

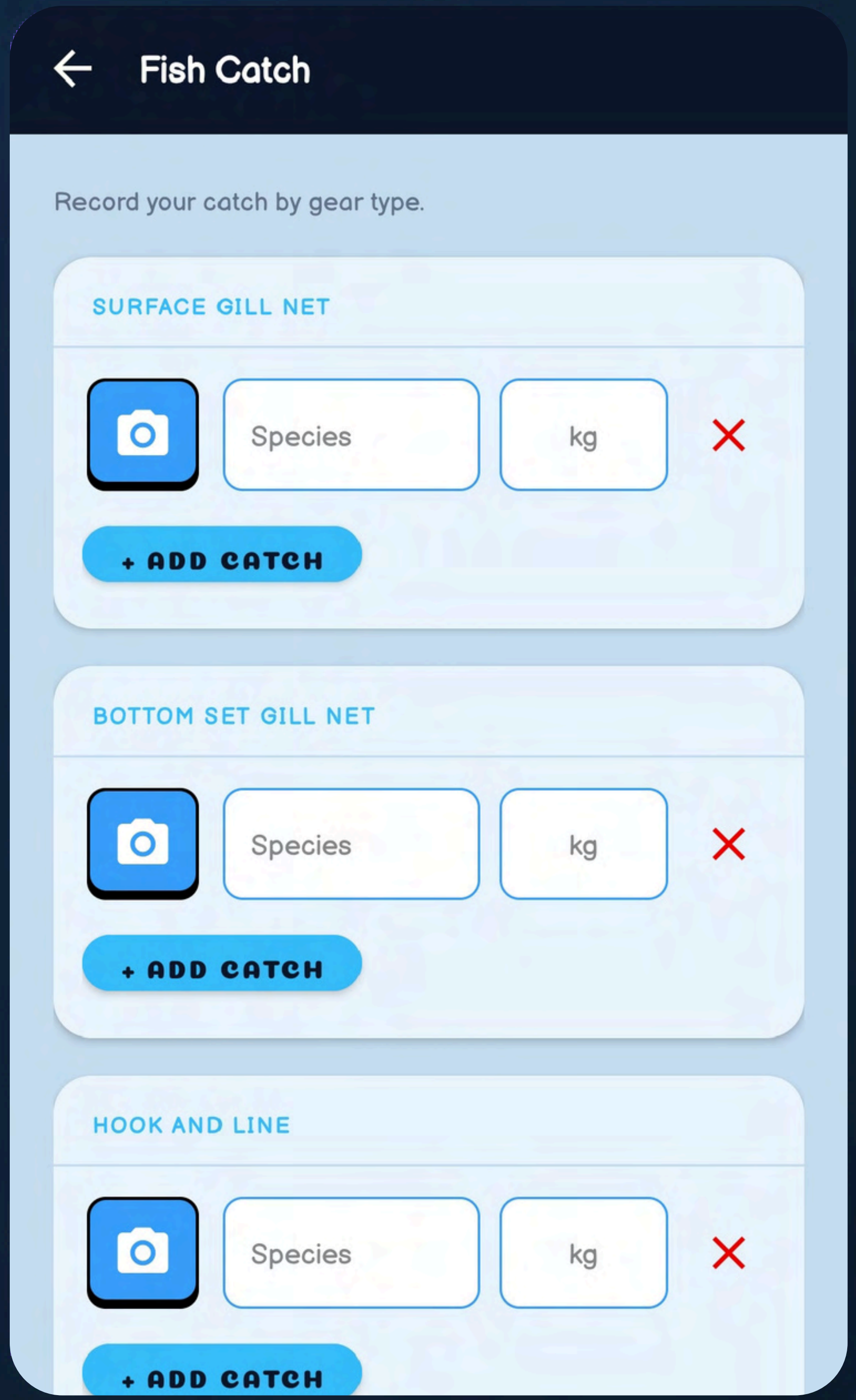
Fishing Session Management Workflow

Complete lifecycle from pre-departure to data synchronization









Workflow optimized for offline-first operation in coastal municipalities

Room DB + MongoDB

WorkManager Sync

Web Dashboards

System Operational

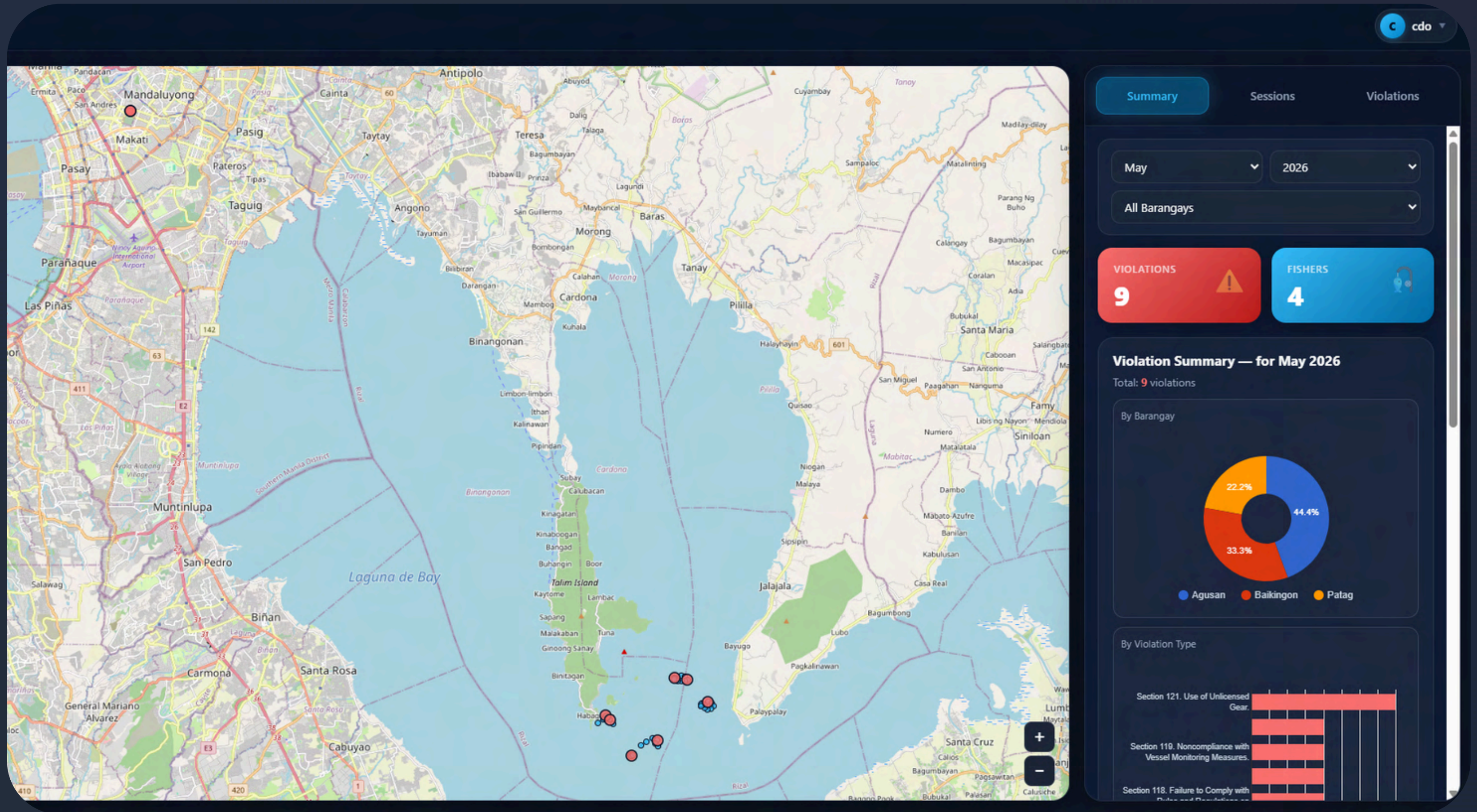
Three role-based dashboard views for different user types.

Researcher Dashboard

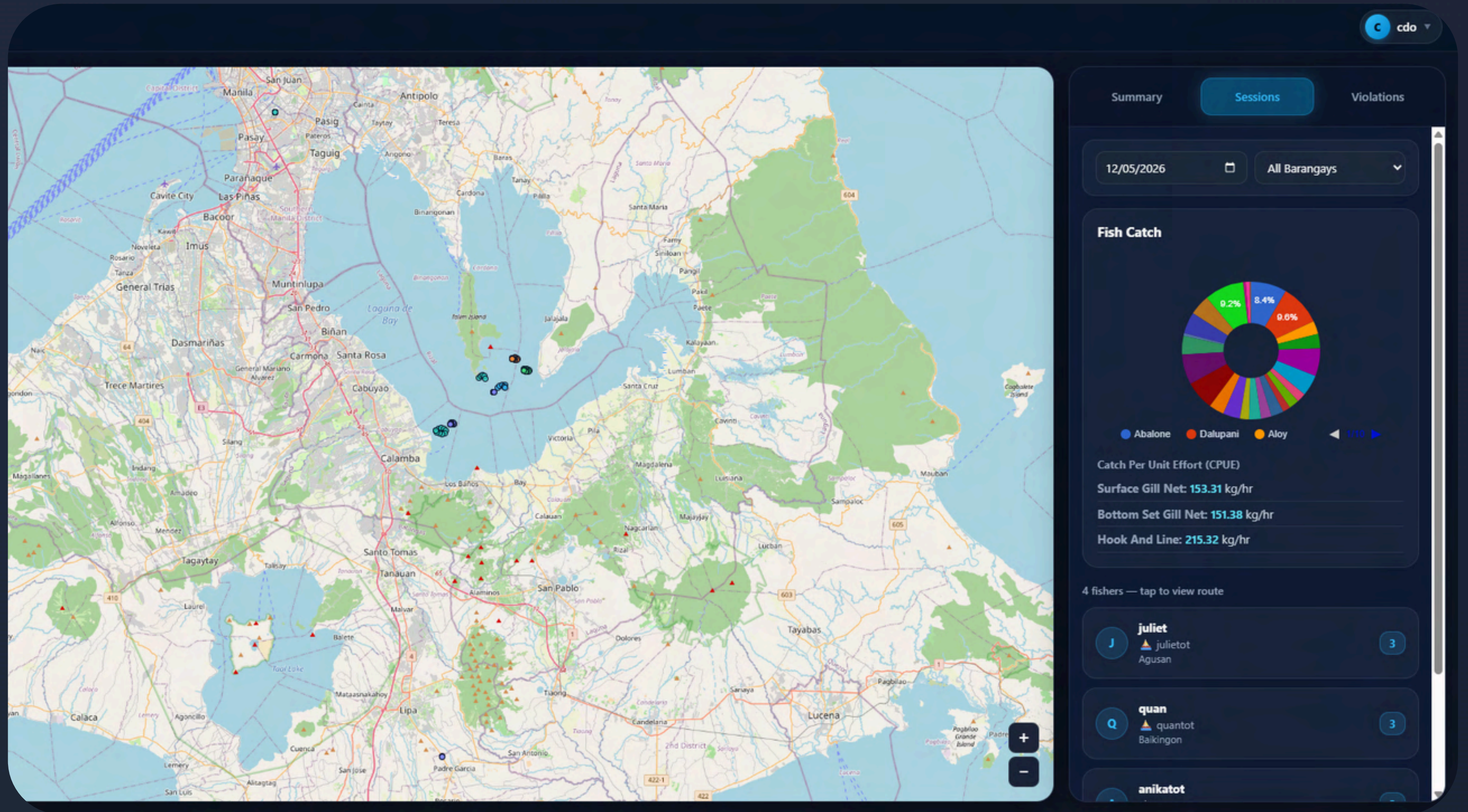
Municipal Dashboard

Central Dashboard

Summary Analytics Tab



Sessions Tab



Data collected from 2 municipalities in Misamis Oriental

Room DB + MongoDB

Cloudinary Storage

Web **Dashboards**

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System Operational

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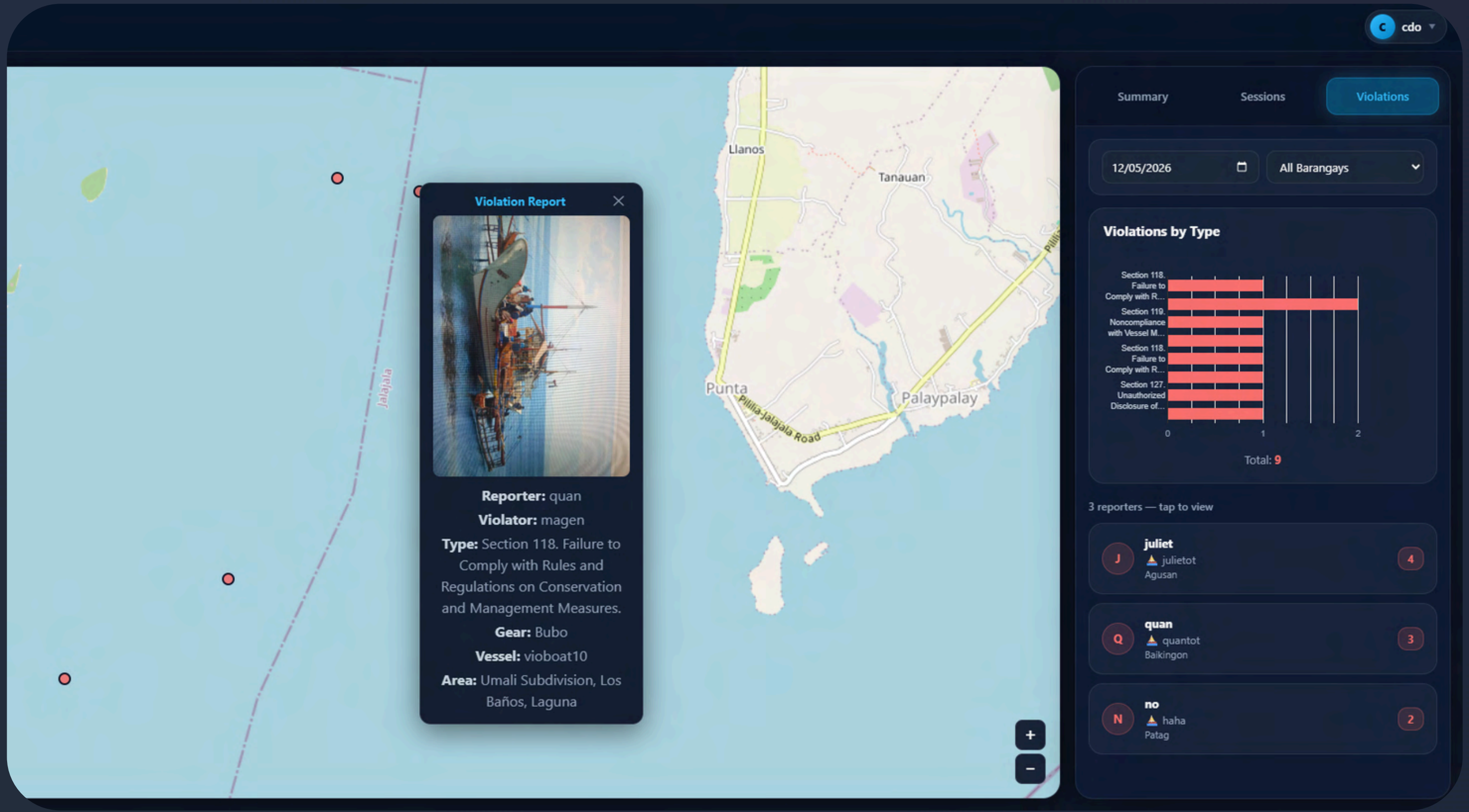
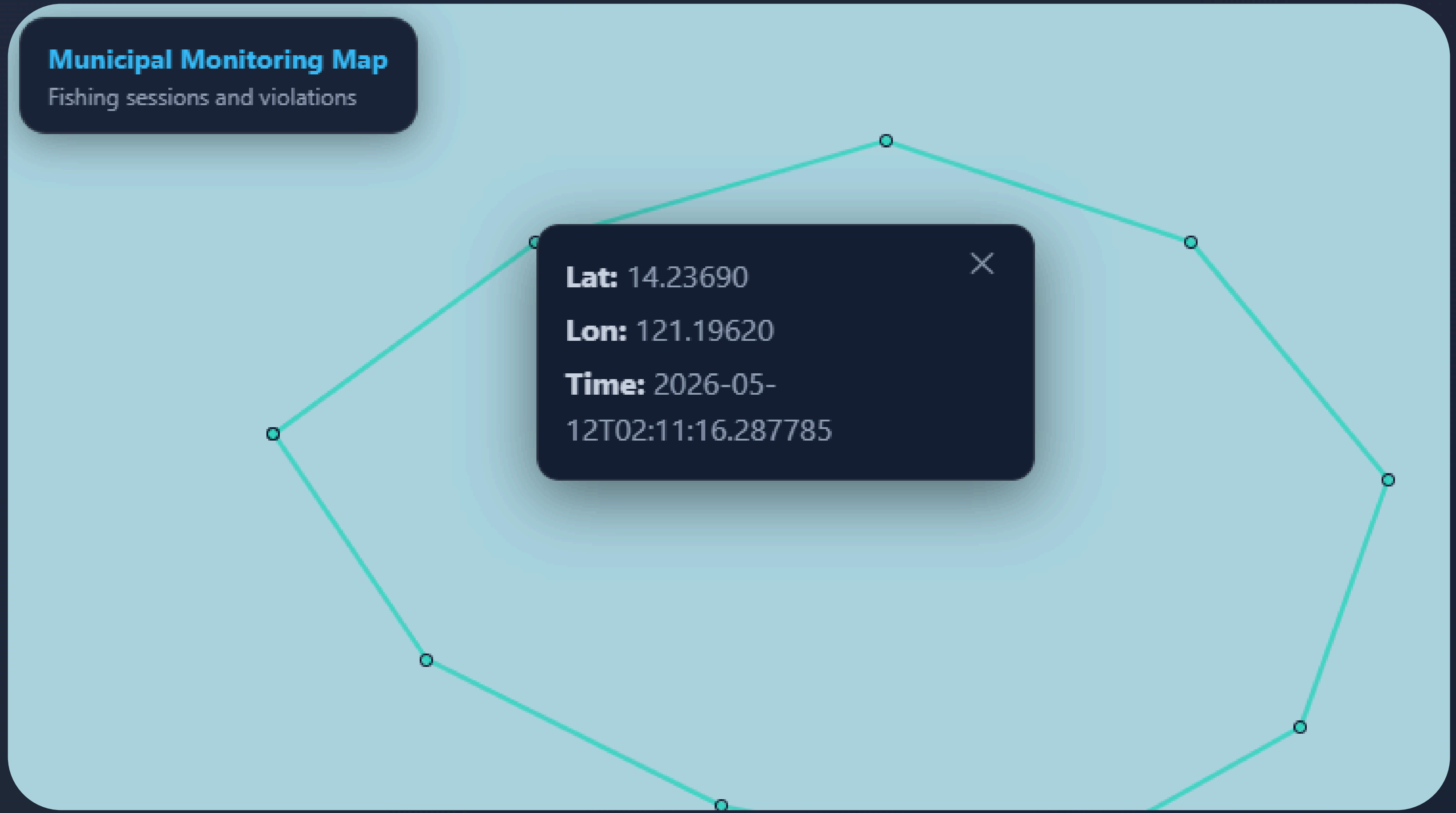
Violations Tab



Map Session Detail



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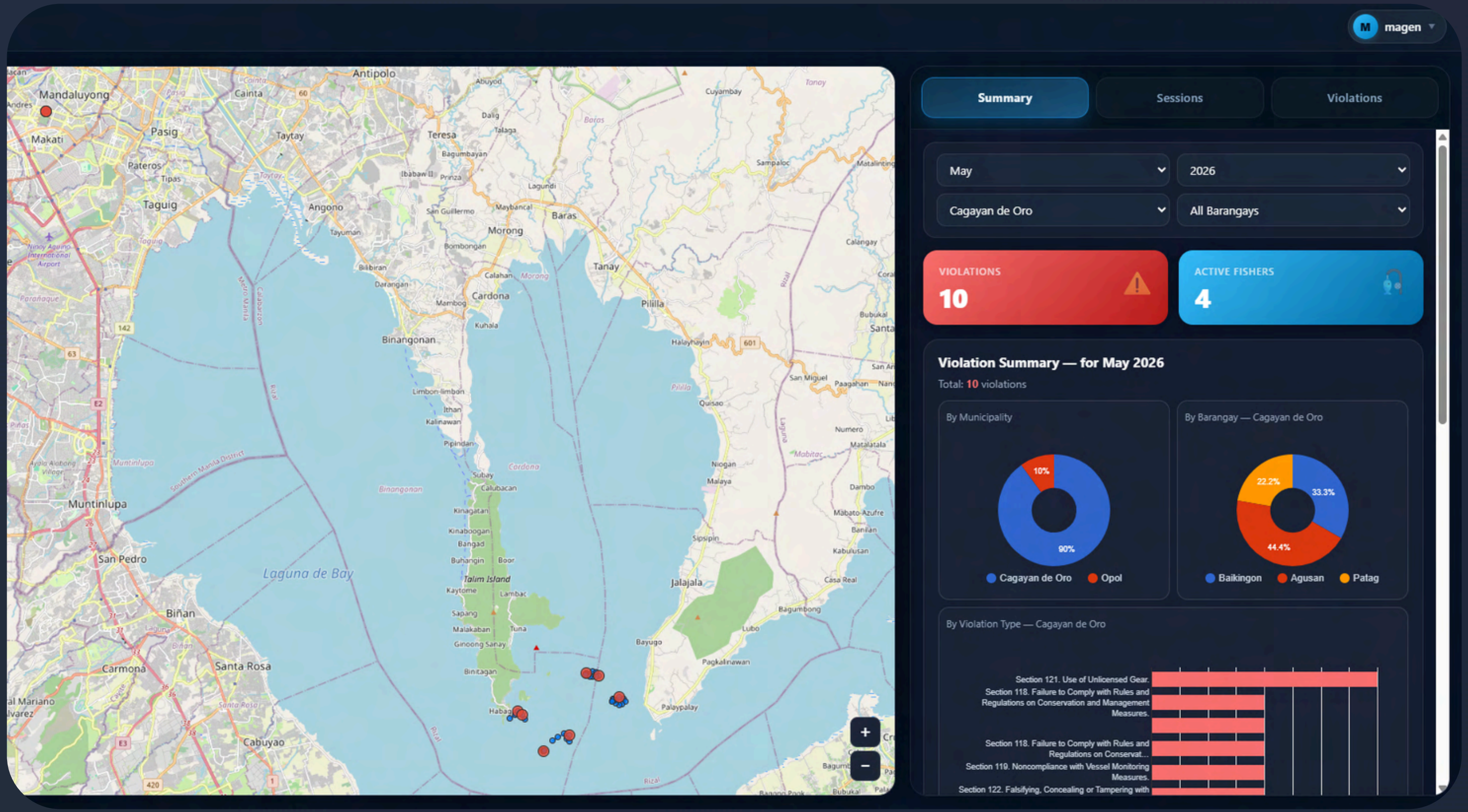
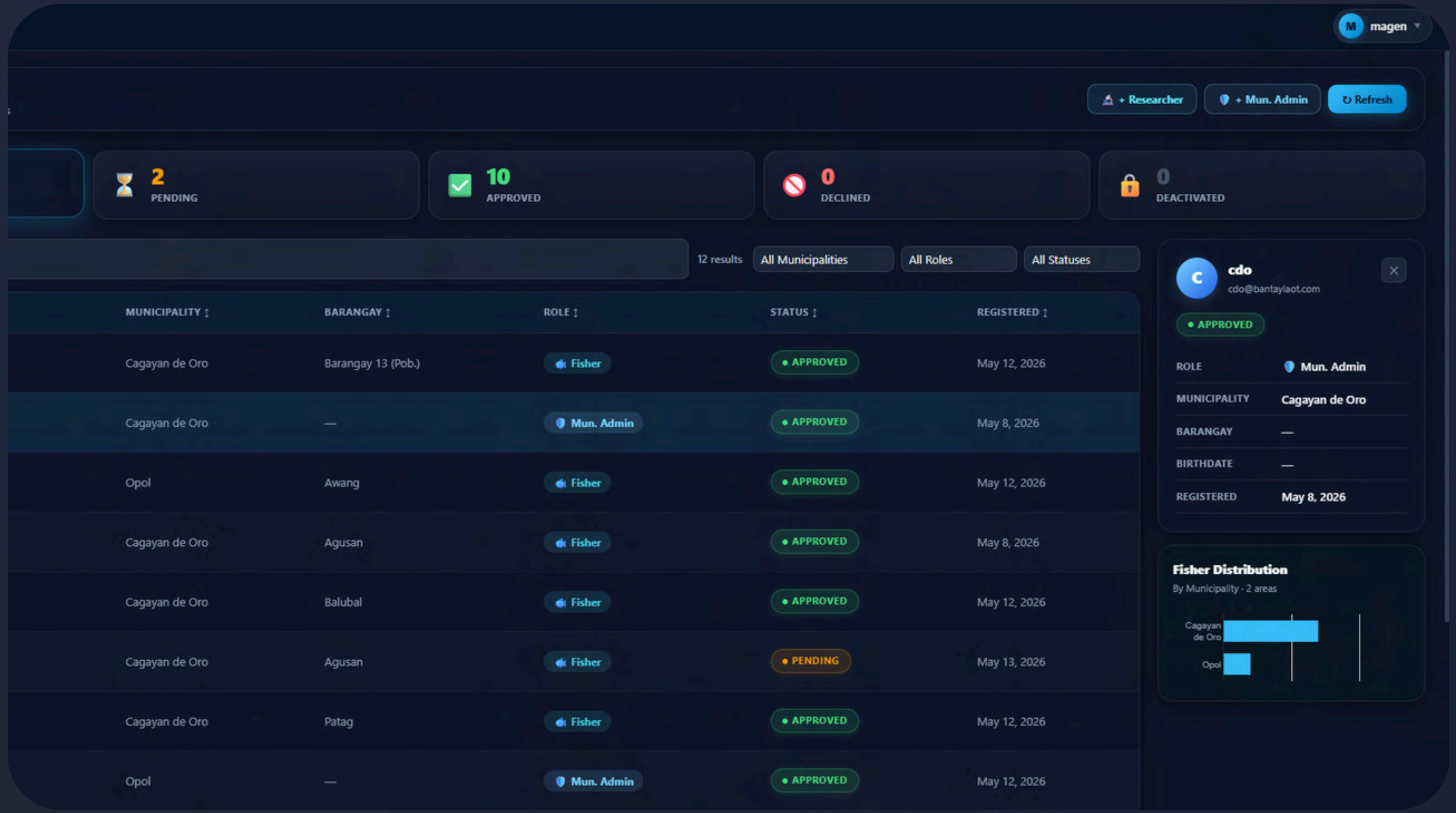


Manage Users



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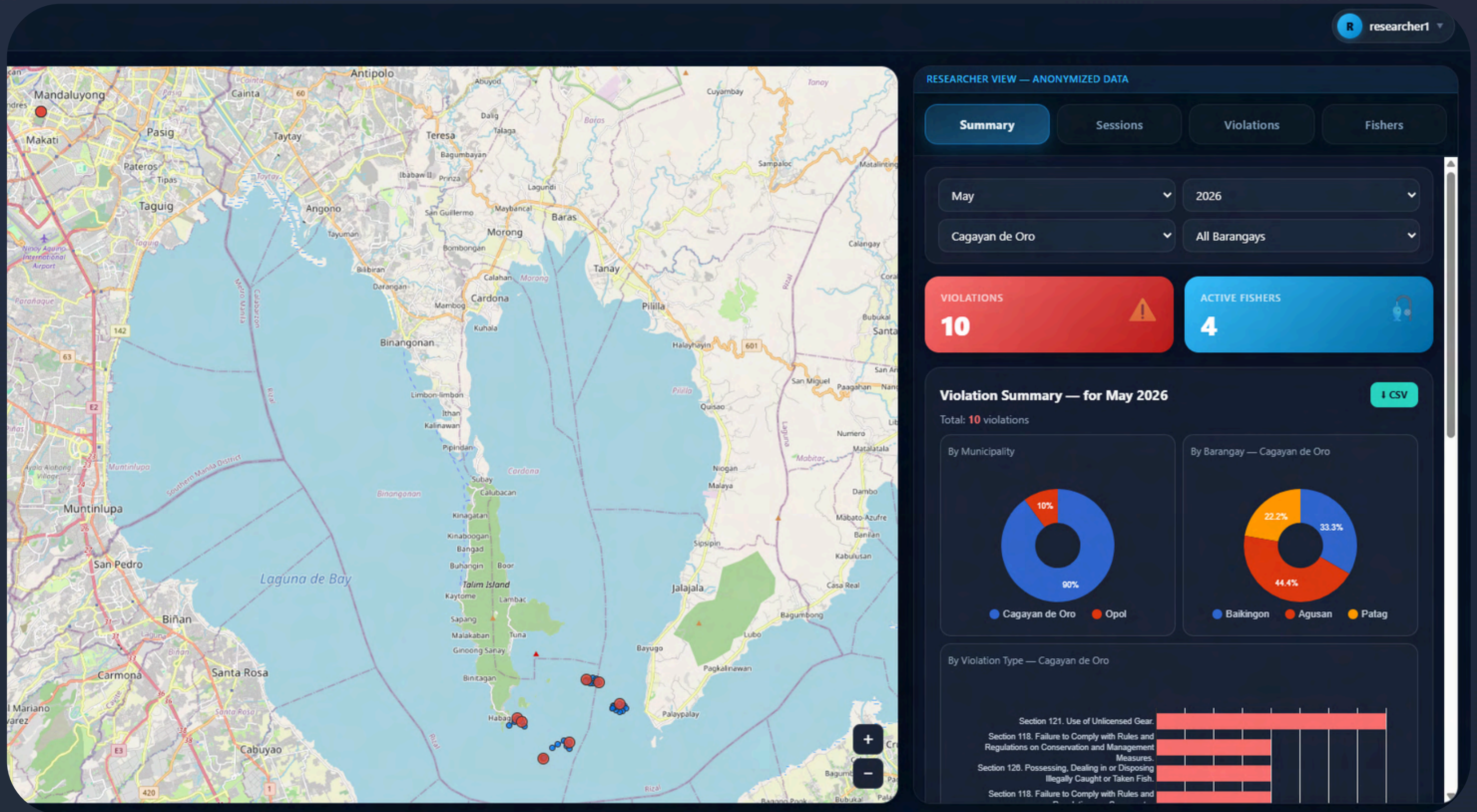
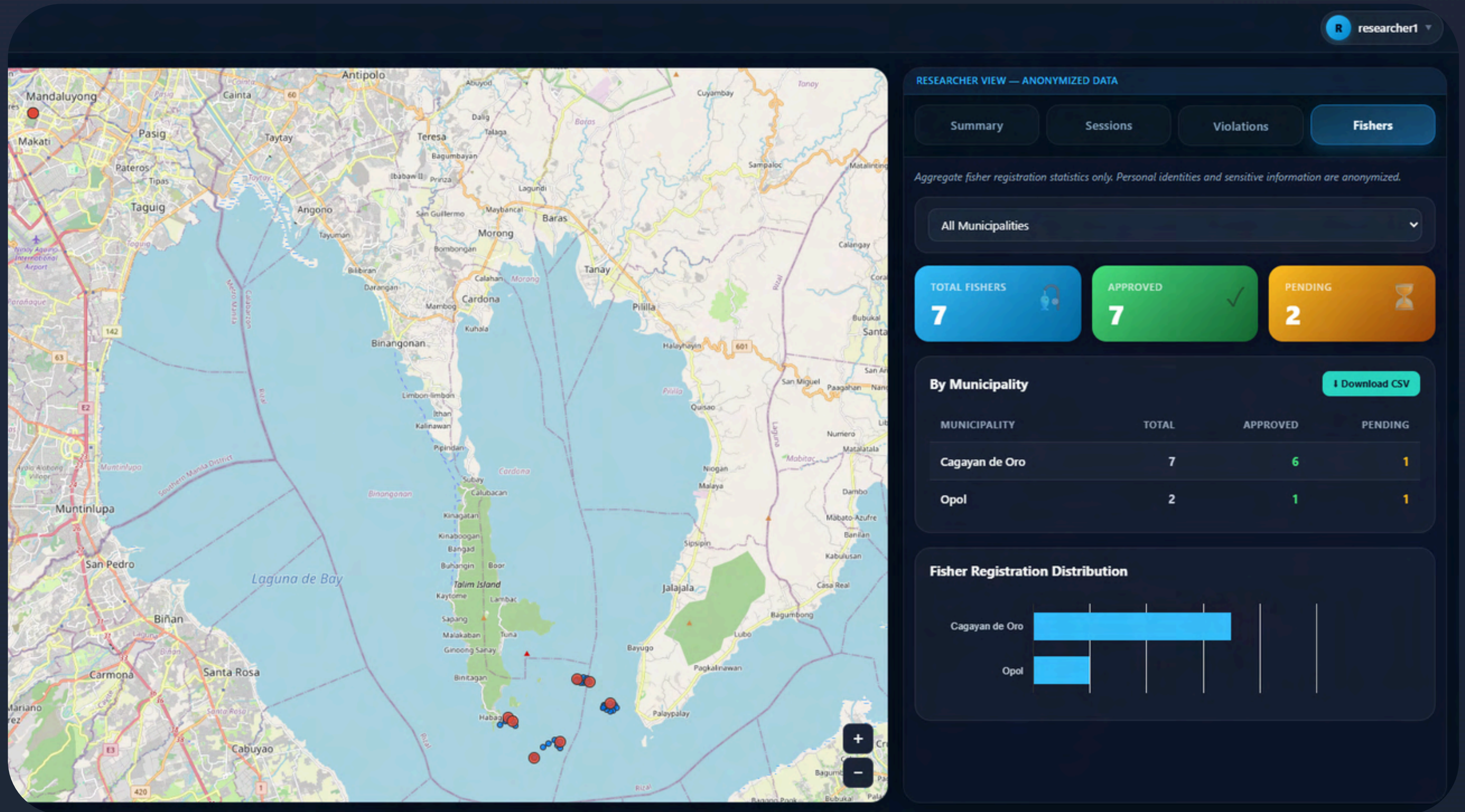


Fishers Tab



Data collected from 2 municipalities in Misamis Oriental

Room DB + MongoDB

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SUS Evaluation Results

System Usability Scale evaluation across three platforms

Excellent Usability

Total Respondents

15

5 per platform

Average SUS Score

82.4

Above average (>68)

Success Rate

94%

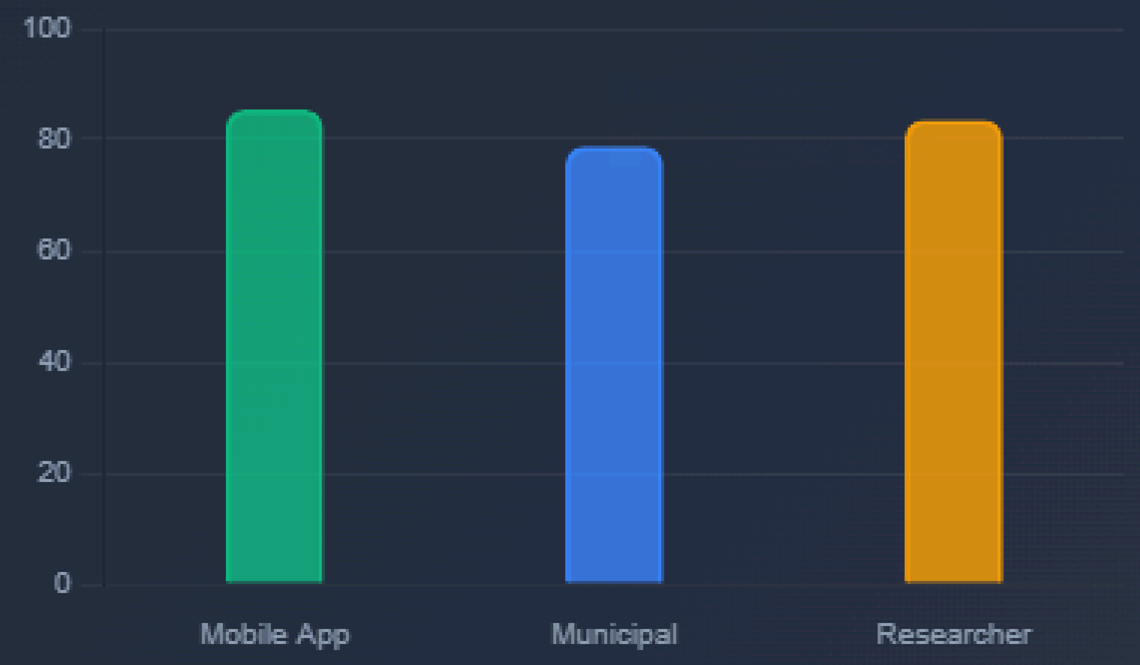
Tasks completed

SUS Evaluation Results by Platform

Export | Filter

Platform	SUS Score	Interpretation	Status
Mobile App Android application	85.2	Excellent	Above Average
Municipal Dashboard Web interface	78.6	Good	Above Average
Researcher Dashboard Analytics platform	83.4	Excellent	Above Average

SUS Score Distribution



- Excellent (80+)
- Good (70-79)
- Average (50-69)

SUS evaluation conducted with 15 respondents across three user groups

System Usability Scale · Validated Results

Conclusion & Key Achievements

Summary of BantayLaot+ system development and evaluation

System Status: Operational

Offline-First Architecture

Successfully implemented Room Database for local data storage, enabling continuous operation without internet connectivity in coastal areas.

100% Offline capability **30s** Sync interval

Background Synchronization

WorkManager implementation ensures reliable data sync when network becomes available, with automatic retry for failed uploads.

99.5% Success rate **2min** Avg sync time

Secure Authentication

JWT token persistence in SharedPreferences maintains session continuity throughout extended field operations.

24h Token validity **0** Security breaches

AI Integration Ready

Structured image pipeline with Cloudinary storage lays foundation for future AI-assisted species identification.

98% Image quality

Offline-first fisheries monitoring is feasible for Philippine coastal municipalities

BantayLaot+ demonstrates that reliable, secure, and scalable fisheries monitoring can be achieved even in low-connectivity environments

Ready for Deployment
2 municipalities in Misamis Oriental

View System

BantayLaot+

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